Pipes and Cisterns – Formula Sheet

◆ 1. Basic Concept

- Filling pipe → **Positive work (+)**
- Emptying pipe → **Negative work (−)**

◆ 2. Efficiency

If a pipe fills tank in **n hours**:
Efficiency = 1 / n

If a pipe empties tank in **n hours**:
Efficiency = −1 / n

◆ 3. Combined Work

If A fills in x hours and B fills in y hours:
1 hour work = (1/x + 1/y)

Total time = (x × y) / (x + y)

◆ 4. Filling + Emptying

If A fills in x hours and B empties in y hours:
1 hour work = (1/x − 1/y)

Total time = (x × y) / (y − x)
☞ (Only when filling is faster than emptying)

◆ 5. LCM Method (Most Important 🚀)

Example:
A fills in 12 hrs, B empties in 18 hrs

Take LCM = 36 units

Efficiency:
A = 36/12 = 3 units/hour
B = 36/18 = −2 units/hour

Net work = 3 − 2 = 1 unit/hour

Total time = 36 hours

◆ 6. Work Done Formula

Work done in t hours = t / total time

Remaining work = 1 − (t / total time)

◆ 7. Multiple Pipes

1 hour work = (1/x + 1/y − 1/z …)

(Add filling, subtract emptying)

◆ 8. Leak Problems

Leak = emptying pipe

Net work = Filling − Leak

◆ 9. Alternate Opening

If pipes open/close at intervals:

- Calculate work step by step
- Use cycle method

◆ 10. Shortcut Trick

If net work = small value → time will be large